

Glucose metabolism inhibits apoptosis in neurons and cancer cells by redox inactivation of cytochrome *c*

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Neurons and cancer cells use glucose extensively, yet the precise advantage of this adaptation remains unclear. These two seemingly disparate cell types also show an increased regulation of the apoptotic pathway, which allows for their long-term survival¹. Here we show that both neurons and cancer cells strictly inhibit cytochrome *c*-mediated apoptosis by a mechanism dependent on glucose metabolism. We report that the pro-apoptotic activity of cytochrome *c* is influenced by its redox state and that increases in reactive oxygen species (ROS) following an apoptotic insult lead to the oxidation and activation of cytochrome *c*. In healthy neurons and cancer cells, however, cytochrome *c* is reduced and held inactive by intracellular glutathione (GSH), generated as a result of glucose metabolism by the pentose phosphate pathway. These results uncover a striking similarity in apoptosis regulation between neurons and cancer cells and provide insight into an adaptive advantage offered by the Warburg effect for cancer cell evasion of apoptosis and for long-term neuronal survival.

Apoptosis is a genetically regulated process that is essential for the development of the organism, but its dysregulation can lead to neurodegenerative disorders as well as cancer^{2,3}. A crucial event in the apoptotic pathway is the release of cytochrome *c* from the mitochondria. In healthy cells, cytochrome *c* resides in the mitochondrial intermembrane space where it serves as a redox carrier for the electron transport chain. However, in response to many apoptotic stimuli, cytochrome *c* is released into the cytosol where it can initiate the formation of the apoptosome complex, leading to caspase activation and subsequent cell death⁴. Emerging evidence indicates that cells, such as postmitotic neurons, that last the lifetime of the organism, as well as cancer cells, which must overcome a cell death response, inhibit the apoptotic pathway¹. Interestingly, despite the striking morphological and functional differences between neurons and cancer cells, both extensively metabolize glucose^{5,6}. Here we examined whether the reliance on glucose metabolism was important for the increased resistance to apoptosis in neurons and cancer cells.

We assessed the ability of endogenous cytochrome *c* to activate apoptosis in sympathetic neurons by using a truncated form of the pro-apoptotic protein Bid (tBid). Full-length Bid is cleaved intracellularly into tBid in response to certain apoptotic stimuli, where it then acts as a potent inducer of cytochrome *c* release from mitochondria⁷. Expression of tBid-GFP plasmid DNA in mouse embryonic fibroblasts (MEFs) resulted in rapid and complete apoptosis, but sympathetic neurons maintained in nerve growth factor (NGF) were resistant to expression of tBid-GFP (Fig. 1a, Supplementary Information, Fig. S1). Sympathetic neurons are known to be resistant to microinjections of exogenous cytochrome *c* because of the inhibition of caspases by XIAP. In contrast to wild-type neurons, XIAP-deficient sympathetic neurons are sensitive to microinjection of excess exogenous cytochrome *c*⁸. However, we were surprised to find that the release of endogenous cytochrome *c* with tBid was incapable of inducing apoptosis even in XIAP-deficient neurons (Fig. 1b). Although it did not induce apoptosis, tBid was completely capable of releasing cytochrome *c* in these neurons (Fig. 1c; Supplementary Information, Fig. S2a). Similarly, injection of a Bid BH3 peptide in neurons induced the release of cytochrome *c* from mitochondria but this did not result in death (Supplementary Information, Figs S1c, S2b). Interestingly, despite having released cytochrome *c*, these neurons maintain their mitochondrial membrane potential (Supplementary Information, Fig. S2c)⁹. Thus, in NGF-maintained neurons, the release of endogenous cytochrome *c* was incapable of inducing apoptosis even in the absence of XIAP (Supplementary Information, Fig. S3). To focus on this unexpected XIAP-independent mechanism of post-cytochrome *c* regulation, we conducted all following experiments in neurons isolated from XIAP-deficient mice.

Despite the resistance of neurons to death following cytochrome *c* release by tBid, sympathetic neurons readily undergo cytochrome *c*-mediated cell death in response to multiple apoptotic stimuli^{10,11}. Thus, apoptotic stimuli must render endogenous cytochrome *c* capable of activating apoptosis in neurons. Indeed, NGF deprivation, as well as DNA damage with etoposide, sensitized sympathetic neurons to apoptosis induced by tBid-mediated cytochrome *c* release (Fig. 1d; Supplementary Information, Figs S1b, c, S4a). As tBid may cause the release of multiple

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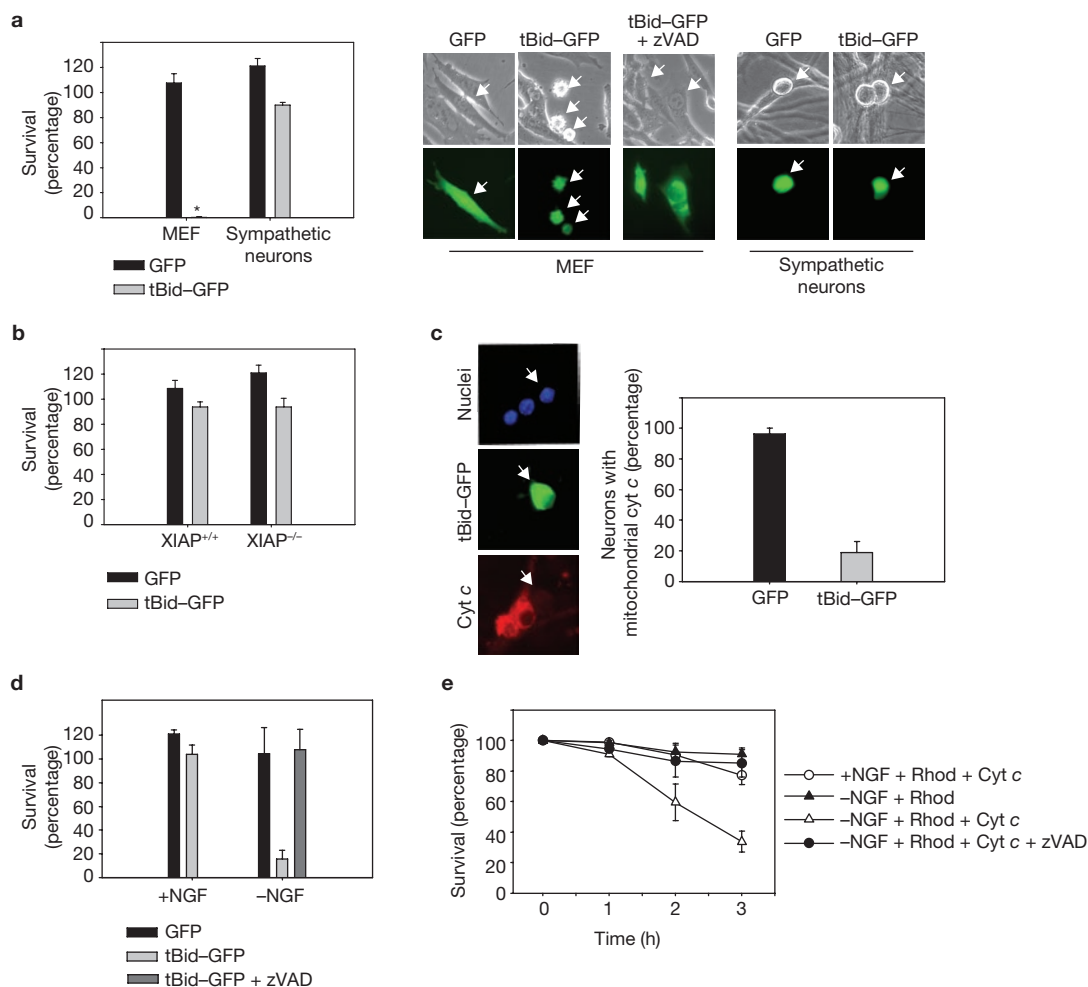


Figure 1 Endogenous cytochrome *c* release is incapable of inducing apoptosis in NGF-maintained sympathetic neurons. **(a)** Cultures of MEFs or sympathetic neurons were injected with tBid-GFP or GFP plasmid. Cell survival was quantified by cell morphology and expressed as a percentage of healthy green cells at 24 h, compared with 5 h post-injection (*no MEF survival). Representative photographs of MEFs and neurons are shown; arrows indicate injected cells. **(b)** Sympathetic neurons from XIAP^{-/-} or wild-type (XIAP^{+/+}) mice were injected with GFP or tBid-GFP, and cell survival was quantified as in **a**. **(c)** tBid-GFP was injected into sympathetic neurons, and allowed to be expressed over 24 h. Cells were fixed and the status of cytochrome *c* analysed by immunofluorescence

microscopy. Percentage of neurons with mitochondrial cytochrome *c* was quantified in XIAP^{-/-} neurons. **(d)** XIAP^{-/-} sympathetic neurons were either maintained in NGF-containing medium, or deprived of NGF for 8 h (with or without 50 μ M zVAD), followed by injection with GFP or tBid-GFP. Cell survival was quantified by cell morphology and expressed as a percentage of healthy green cells at 16 h, compared with 5 h post-injection. **(e)** XIAP^{-/-} sympathetic neurons were deprived of NGF (with or without 50 μ M zVAD) for 12 h followed by injection with rhodamine alone, or cytochrome *c* (2.5 μ g μ l⁻¹) and rhodamine to label injected cells. Cell survival was assessed at various time points following injection. Data are mean $\pm$ s.e.m. of three independent experiments.

factors from the mitochondria, we examined whether this observed effect was a direct result of sensitization to cytochrome *c*. Indeed, NGF-deprived or DNA-damaged neurons injected with cytochrome *c* (at a time-point before endogenous cytochrome *c* release) also showed increased sensitivity (Fig. 1e; Supplementary Information, Fig. S4b).

To be apoptotically active, cytochrome *c* must exist as a holoenzyme complete with its haem prosthetic group¹². *In vitro* studies have examined whether the redox state of cytochrome *c* affects its apoptotic activity. Some studies show that oxidized cytochrome *c* is more apoptotically active, but others suggest that the reduced form is also functional^{13–19}. In particular, a recent study showed that oxidation of cytochrome *c* by cytochrome *c* oxidase promotes caspase activation, whereas its reduction by tertamethylphenylenediamine blocks caspase activation¹⁸. Similarly, oxidized but not reduced cytochrome *c* was found to promote apoptosis in permeabilized HepG2 cells¹⁹. To determine whether the intracellular

redox environment affects the sensitivity of intact neurons to cytochrome *c*, we treated neurons with either low levels of hydrogen peroxide (H₂O₂) to create a more oxidized environment, or cell-permeable reduced glutathione (GSH) to simulate a reduced environment. A high death rate was observed for neurons injected with cytochrome *c* (2.5 μ g μ l⁻¹) only 10–16 h after injection (Fig. 2a, b). However, H₂O₂ greatly increased the sensitivity of neurons to injected cytochrome *c* (Fig. 2a), whereas neurons treated with GSH were resistant (Fig. 2b). These results show that in intact cells, the redox environment has a marked effect on the ability of cytochrome *c* to promote apoptosis.

As changes in the redox environment can have widespread effects, we determined whether the ability of the reduced cellular environment to inhibit cytochrome *c*-mediated apoptosis is a result of the specific inactivation of cytochrome *c*. Incubation of exogenous cytochrome *c* (which is predominantly oxidized) with cytochrome

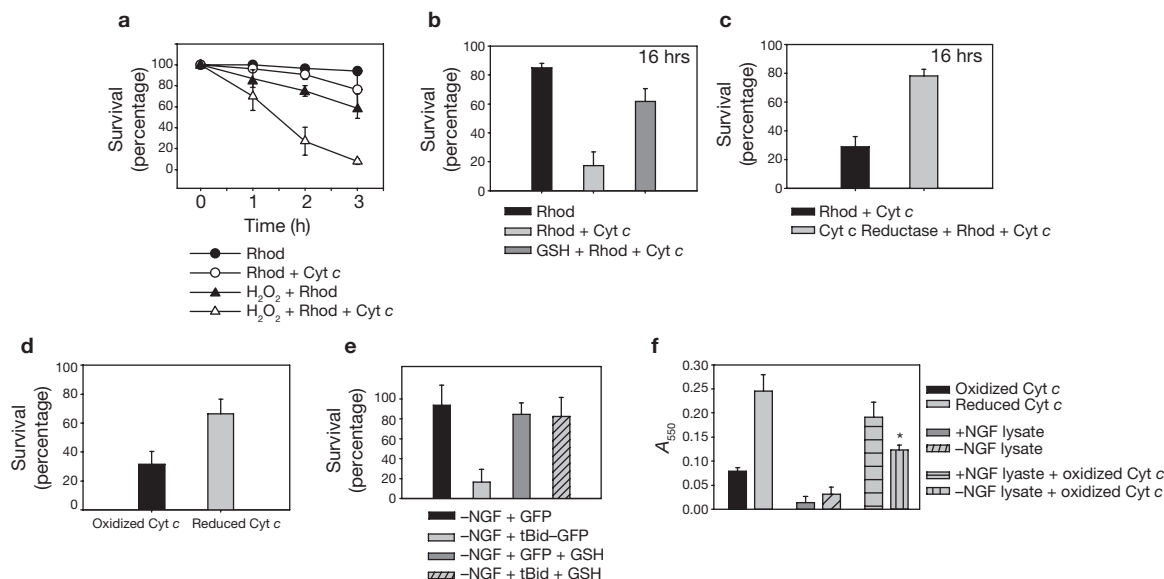


Figure 2 Oxidation of cytochrome *c* increases its apoptotic activity. (a) *XIAP*^{-/-} sympathetic neurons were treated with H_2O_2 (20–50 μ M) for 20 min, then injected with cytochrome *c* protein and rhodamine, or rhodamine dye alone. Cell survival was assessed at various time points after injection. (b) *XIAP*^{-/-} sympathetic neurons were treated with GSH ethyl ester (10 mM) for 12 h, then injected cytochrome *c* and rhodamine, or rhodamine alone. Cell survival was assessed at 16 h after injection. (c) *XIAP*^{-/-} sympathetic neurons were injected with either cytochrome *c* or cytochrome *c* that had been pre-incubated with cytochrome *c* reductase (10 U ml⁻¹). Cell survival was assessed 16 h after injection. (d) *XIAP*^{-/-} sympathetic neurons were injected with either cytochrome *c*, or cytochrome *c* that had

been pre-incubated with DTT (and subsequently separated) to reduce this cytochrome *c*. Percent survival was assessed after 6 h. (e) *XIAP*^{-/-} sympathetic neurons were deprived of NGF for 8 h in the presence of GSH (10 mM), then injected with GFP or tBid-GFP constructs. Cell survival was quantified by cell morphology and expressed as a percentage of healthy green cells at 16 h, compared with 5 h post-injection. (f) Exogenous cytochrome *c* (which is primarily oxidized) was added at a concentration of 10 μ M to neuronal extracts, and absorbance (A_{550}) was measured after 15 min incubation. Control experiments measuring absorbance of reduced or oxidized cytochrome *c* are also shown. Data are mean $\pm$ s.e.m. of three independent experiments (* P = 0.0265, paired *t*-test).

c reductase or dithiothreitol rendered this now reduced cytochrome *c* (Supplementary Information, Fig. S7a) less effective in promoting apoptosis of injected neurons (Fig. 2c, d). Thus, oxidized cytochrome *c* was a more potent inducer of apoptosis than reduced cytochrome *c*. In addition, we generated an N52I mutant of cytochrome *c*, which has increased stability²⁰. The redox status of this protein in intact cells is unclear; however, injection of the N52I mutant cytochrome *c* into sympathetic neurons induced significantly less apoptosis than did wild-type cytochrome *c* (Supplementary Information, Fig. S5).

Two predictions may be made on the basis of the observation that cytosolic cytochrome *c* requires an oxidized cellular environment to be pro-apoptotic. First, the ability of tBid-induced cytochrome *c* release to promote apoptosis in MEFs but not in NGF-maintained neurons may be due to a marked difference in their redox environment. Consistent with this, we found that the average intensity of the redox-sensitive dye 5-(and-6)-chloromethyl-2', 7'-dichlorodihydrofluorescein diacetate (CM-H₂ DCFDA) was 5-fold less in sympathetic neurons, compared with MEFs, indicating reduced levels of reactive oxygen species (ROS) in sympathetic neurons (Supplementary Information, Fig. S6a). Second, NGF deprivation, which sensitizes neurons to cytochrome *c*-induced death, may result in an increased oxidized cellular environment. It is known that the cellular redox status of cells can change during apoptosis²¹. Indeed, ROS intensity, as measured by CM-H₂ DCFDA and dihydrorhodamine (DHR), was increased more than 2.5-fold after NGF deprivation in neurons (Supplementary Information, Fig. S6b, c)^{22,23}. This increase in ROS following NGF deprivation was important to permit

cytochrome *c*-mediated apoptosis, as addition of GSH completely inhibited the ability of NGF deprivation to sensitize neurons to cytochrome *c* release by tBid (Fig. 2e). Importantly, we examined whether there was a difference in the ability of NGF-maintained versus NGF-deprived lysates to affect the redox state of cytochrome *c*. As anticipated, although oxidized cytochrome *c* remained oxidized in NGF-deprived lysates, it was rapidly reduced on incubation with NGF-maintained neuronal lysates (Fig. 2f). Similarly, addition of reduced cytochrome *c* resulted in its oxidation in NGF-deprived, but not NGF-maintained neuronal lysates (Supplementary Information, Fig. S7b), indicating that cytosolic cytochrome *c* would exist in a different redox state in NGF-maintained, healthy neurons, compared with those undergoing apoptosis in response to NGF deprivation.

GSH is one of the most prevalent intracellular reducing agents and maintains redox homeostasis by scavenging ROS. We found that endogenous GSH is necessary for maintaining cytochrome *c* in an inactive state in NGF-maintained neurons, as removal of GSH with diethyl maleate (DEM) or inhibition of GSH synthesis by buthionine-sulfoximine (BSO) leads to an increase in ROS²² and sensitizes cells to cytosolic cytochrome *c* (Fig. 3a, b; Supplementary Information, Fig. S8a). Levels of GSH are maintained in cells by reduced nicotinamide adenine dinucleotide phosphate (NADPH), which is generated when glucose is metabolized by the pentose phosphate pathway²⁴. We examined whether the pentose phosphate pathway was important for regulating the redox status of neurons, and thus cytochrome *c*-mediated apoptosis. Neurons treated with the pentose phosphate pathway inhibitors dehydroepiandrosterone

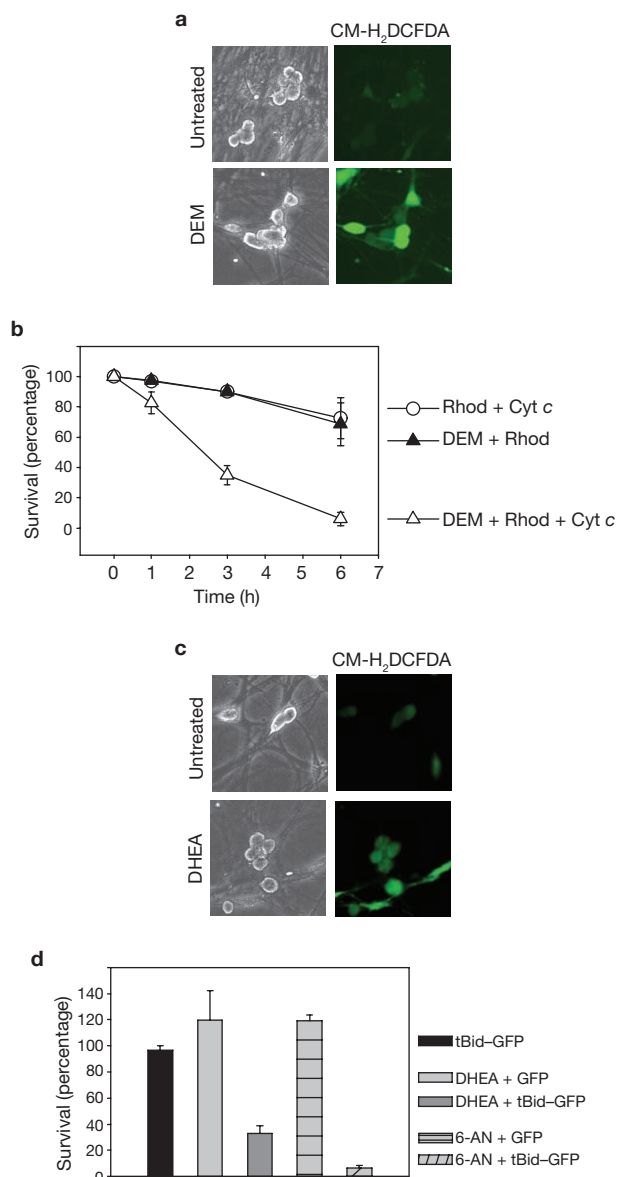


Figure 3 Role of the pentose phosphate shunt in cytochrome *c*-mediated apoptosis. **(a)** Average ROS levels in sympathetic neurons were observed by fluorescence intensity of the redox-sensitive dye CM-H₂DCFDA after 30 min of GSH depletion with DEM (0.1 mM). **(b)** *X/AP*^{-/-} sympathetic neurons (NGF-maintained) were treated with DEM (0.1 mM) for 30 min then injected with cytochrome *c* and rhodamine, or rhodamine dye alone. Cell survival was assessed at various time-points. **(c)** Average ROS levels were observed as in **a** following inhibition of the Pentose Phosphate Pathway with 200 μM DHEA for 24 h. **(d)** *X/AP*^{-/-} sympathetic neurons (NGF-maintained) were treated with DHEA (200 μM) for 6 h, 6-AN (0.1 mM) for 24 h, or left untreated, then injected with tBid-GFP or GFP constructs. Cell survival was expressed as a percentage of healthy green cells at 16 h, compared with 5 h post-injection. Data are mean ± s.e.m. of at least three independent experiments.

(DHEA) or 6-aminocaproic acid (6-AN) accumulated high levels of ROS and became sensitive to endogenous cytochrome *c* release with tBid (Fig. 3c, d; Supplementary Information, Fig. S8b). This effect was not unique to neurons derived from the superior cervical ganglia, as sensory neurons from the dorsal root ganglia (DRG) were also resistant to cytochrome *c* release by tBid, but became sensitive after DHEA treatment (Supplementary Information, Fig. S9). Together,

these data show that glucose metabolism by the pentose phosphate pathway promotes neuronal survival by maintaining GSH levels and directly restricting cytochrome *c*-mediated apoptosis.

As with neurons, many cancer cells are known to metabolize glucose extensively, a phenomenon known as the Warburg effect⁶. To investigate whether increased glucose metabolism in cancer cells directly regulates apoptosis at the level of cytochrome *c*, we first examined the sensitivity of cancer cells to cytosolic injection of cytochrome *c*. Cytochrome *c* was injected in the presence of Smac to eliminate the activity of IAPs that are known to block caspase activation in many cancer cells²⁵, thus focusing on IAP-independent mechanisms of cytochrome *c* regulation. Normal cells readily underwent apoptosis in response to cytosolic cytochrome *c*, whereas cancer cells showed an increased resistance (Fig. 4a; Supplementary Information, Fig. S10a). The decreased sensitivity to cytochrome *c* correlated well with the increased levels of intracellular GSH found in these cancer cells (Fig. 4b). Interestingly, treatment of MEFs with cell-permeable GSH increased their resistance to cytosolic cytochrome *c* (Fig. 4c). In addition, although incubation of oxidized cytochrome *c* with MEF lysate did not significantly affect the redox state of oxidized cytochrome *c*, oxidized cytochrome *c* was rapidly reduced on incubation with lysates generated from cancer cells (Fig. 4d).

We next examined whether enhanced glucose metabolism through the pentose phosphate pathway in HeLa and JM2 cancer cells contributes to cellular redox homeostasis, and thus resistance to cytochrome *c*. Indeed, addition of the pentose phosphate inhibitor DHEA increased ROS levels and specifically rendered these cancer cells sensitive to cytosolic cytochrome *c* (Fig. 4e, f; Supplementary Information, Fig. S10b). Importantly, glucose deprivation also resulted in a similar increase in sensitivity to cytochrome *c* (Fig. 4g). Together, these results show that the ability of cytochrome *c* to induce apoptosis is strictly regulated by its redox state and that glucose flux is a critical regulator of cytochrome *c*-mediated apoptosis.

By coupling the pro-apoptotic activity of cytochrome *c* to the pentose phosphate pathway, cells in which glucose metabolism is increased, such as neurons and cancer cells, are able to effectively maintain a restrictive environment for cytochrome *c*-mediated apoptosis. Such a regulation would be physiologically important for neurons that have limited regenerative potential and survive long-term, and advantageous to cancer cells in evading apoptosis. Flux through the pentose phosphate pathway is likely to inhibit apoptosis under various conditions. For example, upregulation of TIGAR following p53 activation by DNA damage was recently shown to engage the pentose phosphate pathway and promote cell survival²⁶. Overexpression of glucose-6-phosphate dehydrogenase was also shown to promote tumour formation by increasing GSH through pentose phosphate pathway²⁷. Our results suggest that a specific mechanism by which this could inhibit apoptosis is through direct inactivation of cytochrome *c*.

Although maintenance of redox homeostasis is a primary function of the pentose phosphate pathway, flux through this pathway may also inhibit apoptosis at other levels. For example, NADPH production through the pentose phosphate pathway maintains caspase-2 in an inactive state in *Xenopus laevis* egg extracts by a mechanism independent of GSH synthesis²⁸.

Neurons undergoing cytochrome *c*-mediated apoptosis must overcome multiple blocks to undergo cell death. We have shown previously

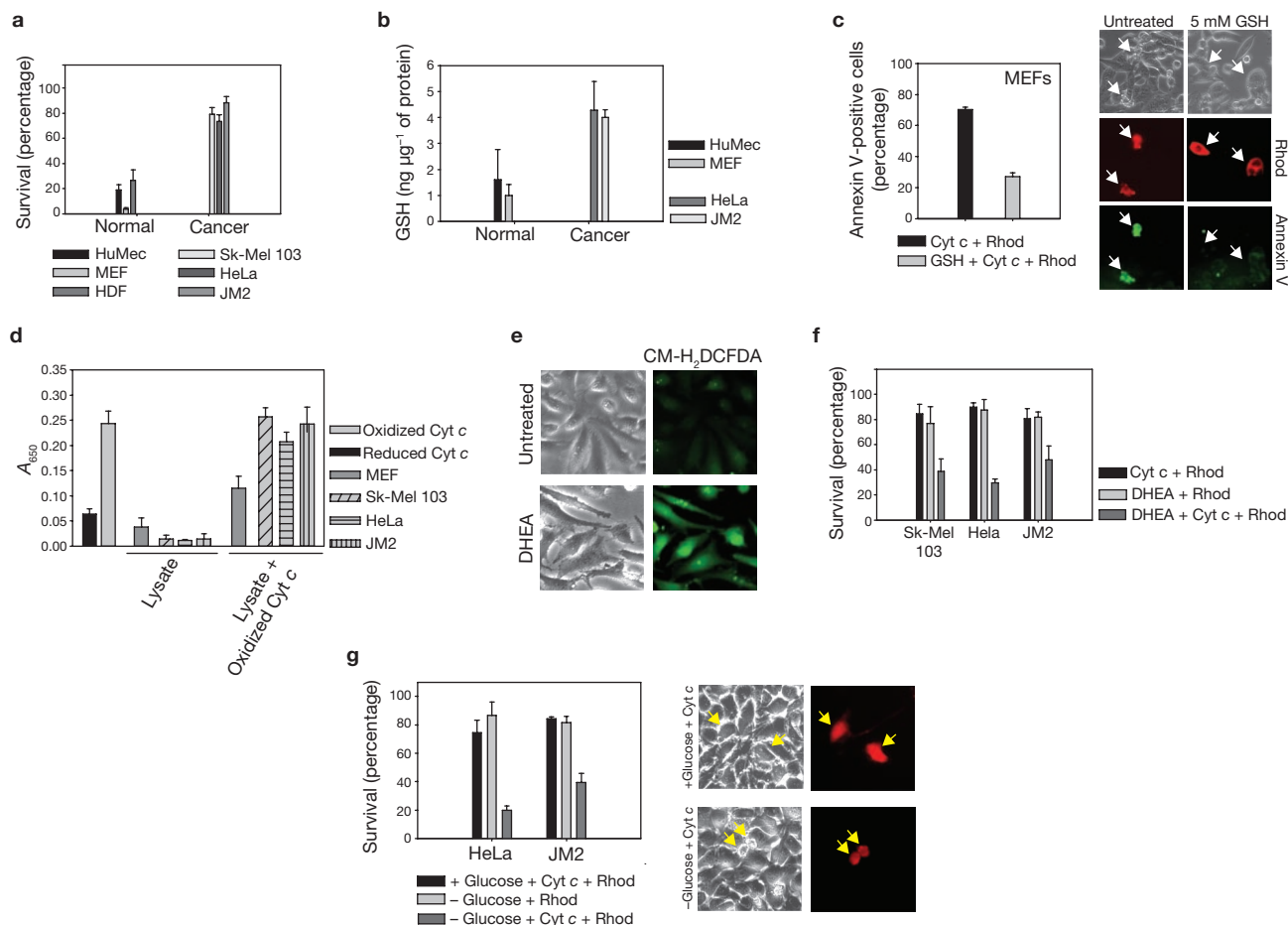


Figure 4 Glucose metabolism protects cancer cells from cytochrome *c*-mediated apoptosis. **(a)** Normal cells (human mammary epithelial cells (HuMECs)), MEFs, human dermal fibroblasts (HDFs) or cancer cell lines (Sk-Mel 103, HeLa, JM2) were injected with cytochrome *c* and Smac protein, and cell survival was assessed after 30 min. **(b)** Total GSH was measured in normal mitotic cells as well as cancer cell lines, and expressed as a concentration of GSH to total cellular protein. **(c)** MEFs were treated with 5 mM GSH ethyl ester for 15 min, followed by injection with cytochrome *c*. After 1 h, injected cells were assessed for annexin V positivity. Photographs show representative annexin V-FITC staining of cytochrome *c* injected cells (arrows). **(d)** Exogenous cytochrome *c* (which is primarily

oxidized) was added at a concentration of 10 μM to normal or cancer cell extract, and absorbance was measured after a 15 min incubation. **(e)** Average ROS levels in HeLa cells measured by fluorescence of CM- H_2DCFDA in the absence or presence of the pentose phosphate pathway inhibitor, DHEA (200 μM) for 6 h. **(f)** The pentose phosphate pathway was inhibited in various cancer cell lines for 6 h by addition of DHEA (200 μM), followed by injection of cytochrome *c* and Smac protein. Cell survival was assessed at 3 h. **(g)** JM2 and HeLa cells were deprived of glucose for 16 h followed by injection with cytochrome *c* and Smac, and assessed for survival at various time points. Images are representative of JM2 cells at 3 h after cytochrome *c* injection. Data are mean $\pm$ s.e.m. of at least three independent experiments.

that endogenous XIAP is a potent inhibitor of cytochrome *c*-mediated apoptosis in neurons⁸, and here, that endogenous cytochrome *c* itself is incapable of inducing apoptosis in NGF-maintained neurons. In response to developmental apoptosis after trophic factor deprivation, these neurons not only showed degradation of XIAP to allow for caspase activation, but also a marked decrease in glucose uptake²⁹ and increase in ROS (Supplementary Information, Fig. S6b, c)^{8,22,23}, which our results show is essential for the activation of cytochrome *c*, and thus, apoptosis.

Increases in intracellular ROS as well as mitochondrial damage are commonly seen during ageing and are associated with apoptotic cell death in many neurodegenerative diseases, including Parkinson's disease^{30,31}. Our results provide mechanistic insight into how even a modest increase in ROS can prime neurons to undergo apoptosis in response to otherwise potentially non-lethal events of mitochondrial damage and cytochrome *c* release.

Neurons and cancer cells are indeed strikingly distinct by most criteria. Different cancer cells themselves appear to inhibit the apoptotic pathway at different points. However, these results bring into focus the possibility that multiple mechanisms evolved by neurons to restrict apoptosis would be similar to those adapted by some mitotic cells during their progression to becoming cancerous. □

METHODS

Reagents. All reagents were purchased from Sigma or Fisher Scientific, unless otherwise stated. Collagenase and trypsin were purchased from Worthington Biochemical Corporation. The GSH:GSSG kit was purchased from Calbiochem. Horse cytochrome *c* was purchased from Sigma. zVAD-fmk was purchased from Enzyme Systems. Annexin V-FITC was purchased from R&D Systems. Recombinant Smac protein was purified from bacteria as described previously⁸ and used at a concentration of 1.5 mg ml^{-1} . MEFs were a gift from Douglas Green (St. Jude's Children's Research Hospital, Memphis, TN). HDFs and HeLa cells were obtained from UNC Tissue Culture Facility, and JM2 cells were a gift from Ekhsan Holmuhamedov (UNC, Chapel Hill,

NC). Sk-Mel 103 cells were a gift from Channing Der (UNC, Chapel Hill, NC). XIAP^{-/-} mice were a gift from Craig Thompson (University of Pennsylvania, Philadelphia, PA).

Sympathetic neuronal cultures. Primary sympathetic neurons were dissected from the superior cervical ganglia of postnatal day 0–1 mice (wild-type ICR or XIAP-deficient C57BL/6 mice) and maintained in culture as described previously⁸. Cells were plated on collagen-coated dishes at a density of 10,000 cells per well. Neurons were grown for 4–5 days in NGF-containing medium before subjecting them to experimental conditions. For NGF deprivation, cultures were rinsed three times with medium lacking NGF, followed by the addition of goat anti-NGF neutralizing antibody. Cells were treated with various compounds: etoposide (20 μ M); DHEA (200 μ M), 6-AN (0.1 mM); DEM (0.1 mM); GSH-ethyl ester (10 mM); BSO (200 μ M).

Microinjection and quantification of cell survival. Cells were injected with horse cytochrome *c* protein (2.5 μ g μ l⁻¹) or Bid BH3 peptide (8 mM) (AnaSpec) and rhodamine dextran (4 μ g μ l⁻¹) in microinjection buffer containing 100 mM KCl and 10 mM KPi, pH 7.4 as described previously⁸. Immediately after injections, the number of rhodamine-positive cells were counted. At various times after injections, the number of viable injected cells remaining was determined using the same counting criteria and expressed as a percentage of the original number of microinjected cells. This method of assessing survival correlates well with other cell-survival assays, such as trypan blue exclusion and staining with calcein AM⁸.

In experiments involving microinjection of DNAs, 100 ng μ l⁻¹ of the tBid expressing plasmid, or EGFP vector in microinjection buffer was injected into the nucleus of neurons or MEFs. To quantify survival after injection, GFP-positive cells were counted 5 and 24 h after injection. Cell survival is expressed as a ratio of viable GFP-positive cells at 24 h post-injection to GFP-positive cells at 5 h post-injection. As tBid rapidly induces death (under certain conditions), the window between detectable expression of tBid–GFP in cells and apoptosis is narrow. Therefore, we chose an early time-point for time zero, to avoid excluding cells that may die soon after this. Under conditions where injection of tBid–GFP or GFP alone does not kill cells, this number becomes greater than 100% as a few additional cells begin to express visible GFP after the initial count. Under conditions where cell death occurred, death was confirmed as caspase-dependent as it could be inhibited with zVAD-fmk (50 μ M). Cell survival was also assessed by annexin-V positivity, according to the manufacturer's protocol.

Immunofluorescence microscopy for assessing cytochrome *c* release. The status of cytochrome *c* (whether intact in the mitochondria or released) was examined by immunofluorescence microscopy. Sympathetic neurons show a punctate mitochondrial staining with anti-cytochrome *c* antibody, which is lost as cytochrome *c* is released from the mitochondria. Briefly, cultured sympathetic neurons were fixed in 4% paraformaldehyde and incubated overnight with anti-cytochrome *c* (1:1000; 556432, BD Biosciences), or anti-GFP (1:100; Cell Signaling) primary antibody followed by a 2 h incubation with anti-mouse Cy3 (1:400), anti-mouse Alexa 488 (1:1000) or anti-chicken Alexa 488 (1:1000) secondary antibody (Jackson Labs). Nuclei were stained with Hoechst 33258 (Molecular Probes).

ROS measurement. The redox-sensitive dye CM-H₂ DCFDA or dihydrorhodamine (DHR) (Molecular Probes) was used to measure ROS levels in sympathetic neuronal cultures and cell lines using the protocol described previously²². CM-H₂ DCFDA is non-fluorescent when reduced and after cell permeabilization, is cleaved by esterases and trapped in the cell where oxidation converts it to a fluorescent form. After treatment of cells with various experimental conditions, cells were incubated with 10 μ M CM-H₂ DCFDA or 10 μ M DHR for 20 min then rinsed three times in PBS. Fluorescent intensity of cells was represented as an image, and/or quantified by measuring pixel intensity in a 50 μ m² area within individual soma using Metamorph software or by measuring the fluorescence intensity using a Fluoroskan (ThermoLabSystems) plate reader.

Preparing reduced cytochrome *c*, and assessing cytochrome *c* redox status. Cytochrome *c* was reduced by incubating it with 10 U ml⁻¹ of cytochrome *c* reductase (NADPH) followed by injection into neurons. Alternatively, cytochrome *c* was

reduced with 0.5 mM DTT, then separated on PD-10 columns (GE Healthcare). The redox status of cytochrome *c* was assessed by monitoring its absorbance at 550 nm on Nanodrop spectrophotometer. For experiments involving analysis of redox activity of cytochrome *c* after NGF deprivation, sympathetic neurons were plated at a density of 100,000 cells and maintained in NGF or deprived of NGF for 12 h. Neuronal lysate was collected and incubated with oxidized cytochrome *c* (10 μ M) for 15 min before analysis, as described above. For mitotic cells, oxidized cytochrome *c* (10 μ M) was incubated with cell lysates (200 μ g) for 15 min before measurement at 550 nm.

Reduced glutathione measurements. Measurements of GSH:GSSG were carried out using the GSH:GSSG ratio kit according to a modified version³² of the manufacturer's. Briefly, cells were lysed in 5% metaphosphoric acid for 15 min, centrifuged and the GSH content of the supernatant was measured by the rate of colorimetric change of 5,5'-dithiobis(nitrobenzoic acid) at 412 nm in the presence of glutathione reductase and NADPH. The acid pellet was resuspended in 1 N NaOH and the solubilized protein concentration measured by the Lowry method (BioRad). Total GSH is represented as ng of GSH per μ g of cellular protein.

Image acquisition and processing. All images were acquired by a Hamamatsu ORCA-ER digital B/W CCD camera mounted on a Leica inverted fluorescence microscope (DMIRE 2). The image acquisition software was Metamorph version 5.0 (Universal Imaging Corporation). Images were scaled down and cropped in Adobe Photoshop to prepare the final figures.

Note: Supplementary Information is available on the Nature Cell Biology website.

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AUTHOR CONTRIBUTIONS

A.E.V. performed all experiments; A.E.V. and M.D. planned the project and analysed the data.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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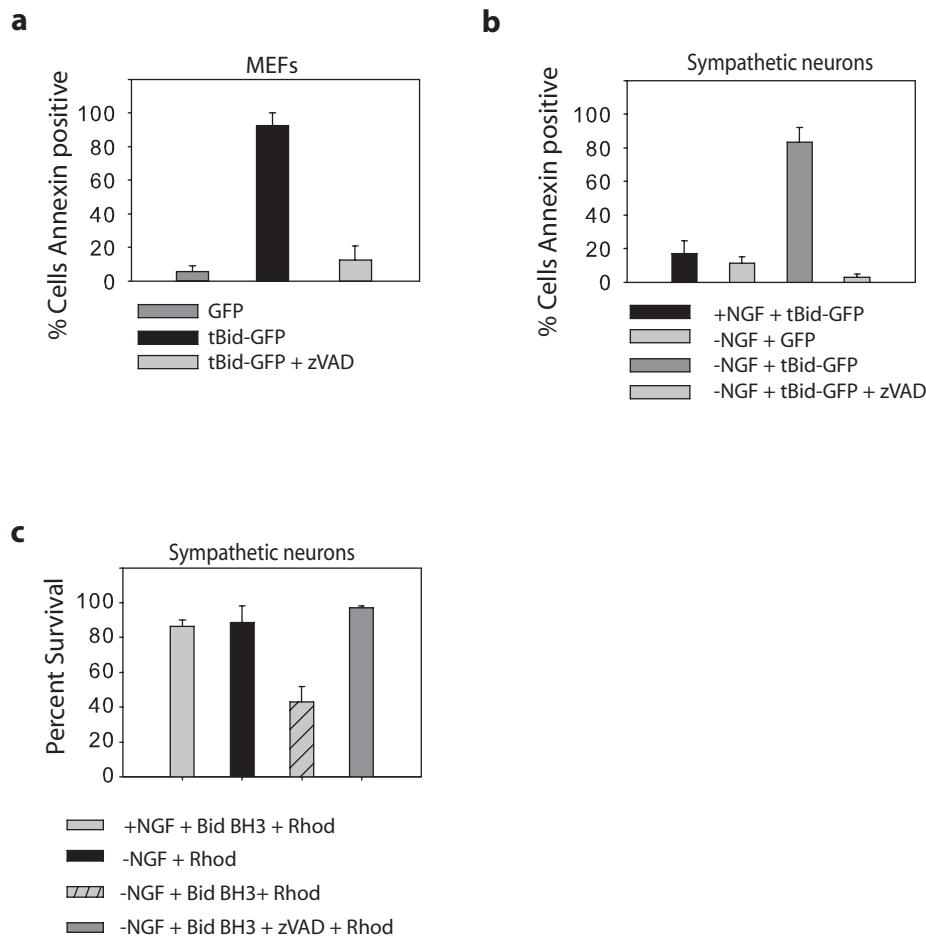


Figure S1 tBid induces an apoptotic cell death in MEFs and NGF-deprived neurons. **a)** MEFs were injected with tBid-GFP in the presence or absence of zVAD. After 5 hrs, cells were analyzed for Annexin V-Cy3 positivity. **b)** XIAP^{-/-} sympathetic neurons were either maintained in, or deprived of NGF for 8 hrs, followed by injection with tBid-GFP in the presence or absence of zVAD. Cells were then analyzed for Annexin V positivity. **c)** NGF deprivation

sensitizes neurons to cytochrome *c* release by Bid BH3. XIAP^{-/-} neurons were either maintained in media containing NGF, or deprived of NGF for 10 hrs, followed by injection with Bid BH3 peptide along with rhodamine to mark the injected cells. zVAD was added to the indicated condition at the time of injection. Cell survival was assessed 6 hrs after injection. Error bars represent $\pm$ SEM of $n \geq 3$.

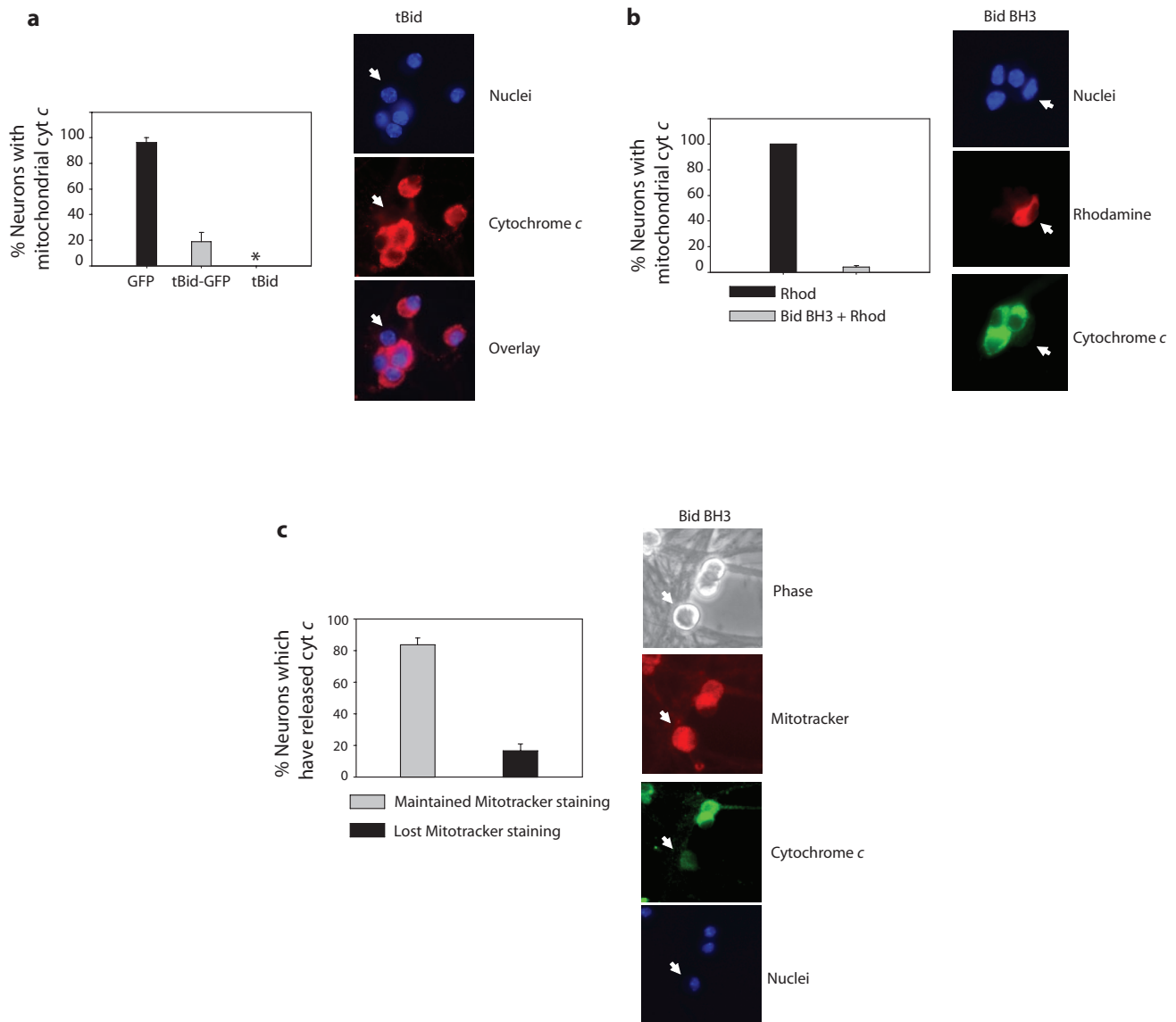


Figure S2 tBid induces cytochrome *c* release but not apoptosis in neurons.

a) XIAP^{-/-} sympathetic neurons were injected with plasmids encoding GFP, tBid-GFP or an untagged tBid construct, and after 24 hrs, the status of cytochrome *c* in injected neurons was assessed and quantified by immunofluorescence. * represents 100% cytochrome *c* release. Photographs are representative of cytochrome *c* staining after 24 hrs of tBid expression. Arrow points to cell injected with tBid. **b)** Neurons were injected with Bid BH3 peptide along with rhodamine to mark the injected cells. Cytochrome *c* status was assessed by immunofluorescence 8 hrs after injection. Arrow points to cell injected with the Bid BH3

peptide. **c)** tBid expressing neurons release cytochrome *c*, but maintain mitochondrial membrane potential. Neurons were injected with Bid BH3 peptide. After 3 hrs, cells were scored either as maintaining membrane potential (Mitotracker staining maintained) or lost membrane potential (Mitotracker staining lost) using Mitotracker Orange. Cells were then fixed and cytochrome *c* release assessed by immunofluorescence. Data is represented as a percentage Bid BH3 injected cells which have released cytochrome *c*, that have either maintained or lost their membrane potential. Arrows represent a cell which has been injected with the Bid BH3 peptide. Error bars represent $\pm$ SEM of $n \geq 3$.

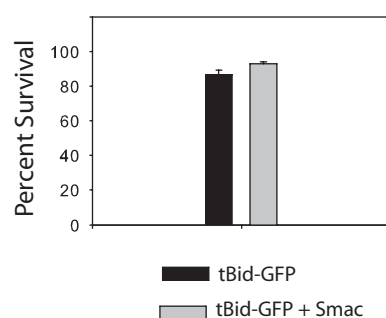


Figure S3 Neuronal resistance to cytochrome *c* release is not mediated by IAPs. XIAP^{-/-} neurons were injected with tBid-GFP, or tBid-GFP + Smac protein. Cell survival was assessed by morphology by quantifying the number of GFP positive cells at 16 hrs compared to 5 hrs post-injection. Error bars represent $\pm$ SEM of $n \geq 3$.

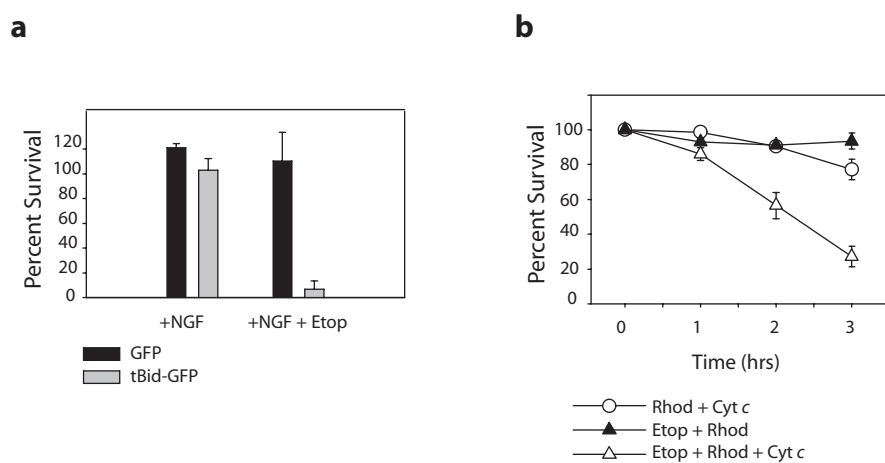


Figure S4 DNA damage sensitizes neurons to cytochrome *c*-mediated apoptosis. **a)** XIAP^{-/-} sympathetic neurons were treated with 20 μ M etoposide for 12 hrs, followed by injection of tBid-GFP or GFP plasmids. Cell survival was assessed by morphology by quantifying the number of viable GFP positive

cells at 16 hrs compared to 5 hrs post-injection. **b)** XIAP^{-/-} sympathetic neurons were treated with 20 μ M etoposide for 12 hrs followed by injection with cytochrome *c* and rhodamine, or rhodamine alone. Cell survival was assessed at multiple time points. Error bars represent $\pm$ SEM of $n \geq 3$

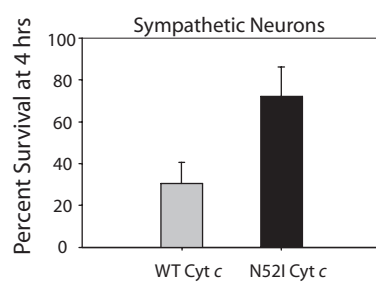


Figure S5 Cytochrome *c* N52I has reduced potency in inducing apoptosis in XIAP^{-/-} intact neurons. XIAP^{-/-} sympathetic neurons were injected with

recombinant wildtype (WT) or N52I cytochrome *c* (4 µg/ul) and assessed for survival after 4 hrs. Error bars represent ±SEM of $n \geq 3$

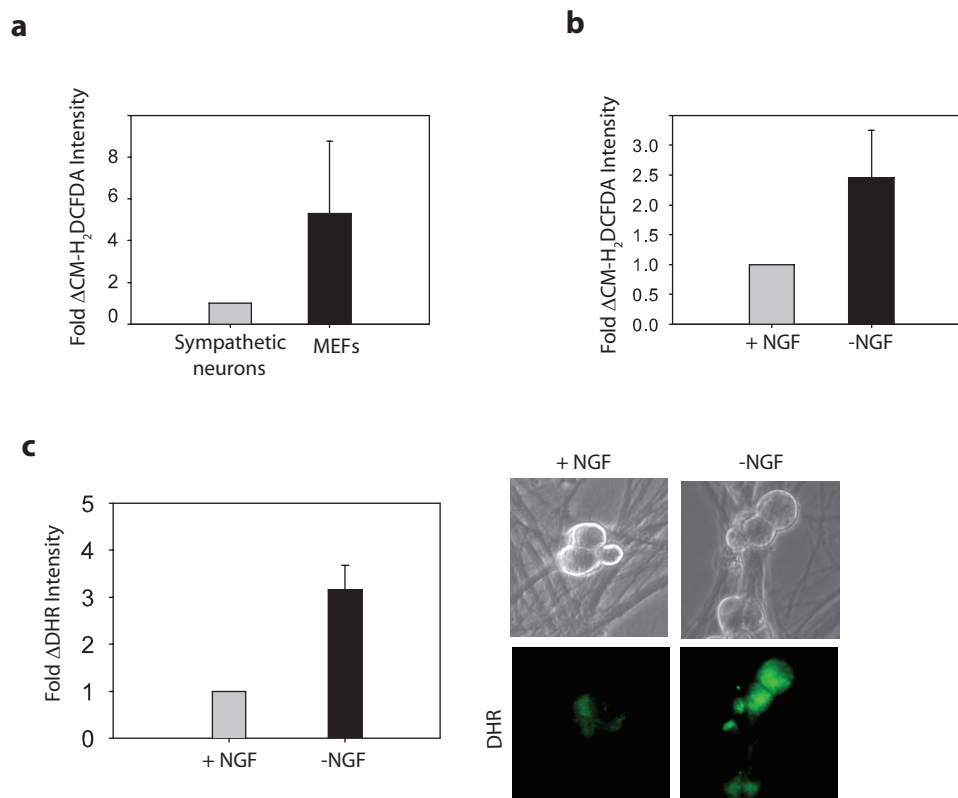


Figure S6 ROS levels in MEFs, and sympathetic neurons after NGF deprivation. **a)** MEFs were co-cultured with sympathetic neurons, followed by incubation with 10 μ M CM-H₂DCFDA redox-sensitive dye. Fluorescent intensity was measured in a 50 μ m² area within individual soma and quantified using Metamorph software. **b)** XIAP^{-/-} sympathetic neurons were

maintained in NGF, or deprived of NGF for 12 hrs, followed by incubation with CM-H₂DCFDA. Fluorescent intensity was quantified as in (a). **c)** Sympathetic neurons were maintained in NGF, or deprived of NGF for 12 hrs, followed by incubation with 10 μ M Dihydrorhodamine (DHR). Fluorescent intensity was quantified as in (a). Representative photographs are shown.

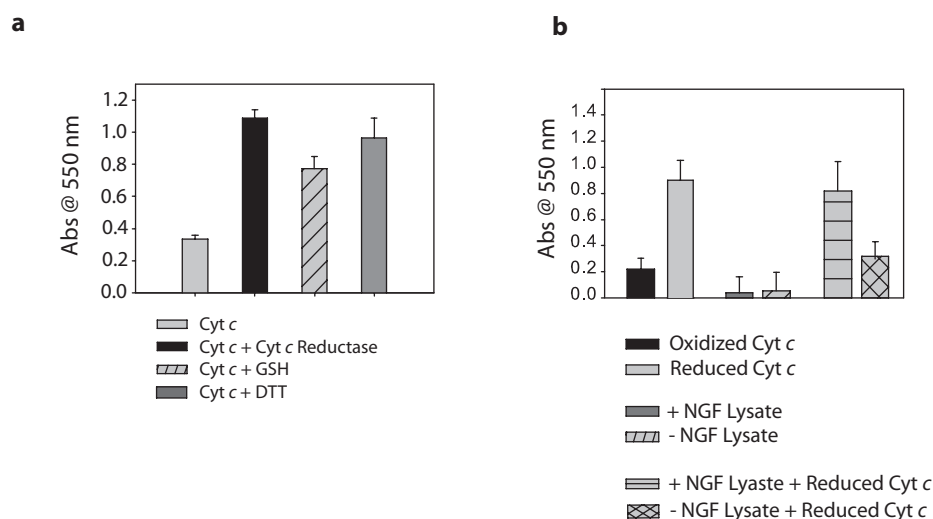


Figure S7 Neurons undergoing apoptosis are able to oxidize cytochrome *c*. **a)** 40 μ M horse cytochrome *c* was incubated with either 10 units/ml cytochrome *c* reductase, 10 mM glutathione ethyl ester (GSH), or 0.5 mM DTT for 15 minutes, followed by measurement of Abs₅₅₀. **b)** Sympathetic neurons were maintained in NGF or deprived of NGF for

12 hrs. Lysates were then incubated with 40 μ M cytochrome *c* which had been reduced with DTT (and subsequently separated from) followed by measurement of Abs₅₅₀. Control experiments measuring Abs₅₅₀ of reduced or oxidized cytochrome *c* are also shown. Error bars represent $\pm$ SEM of $n \geq 3$

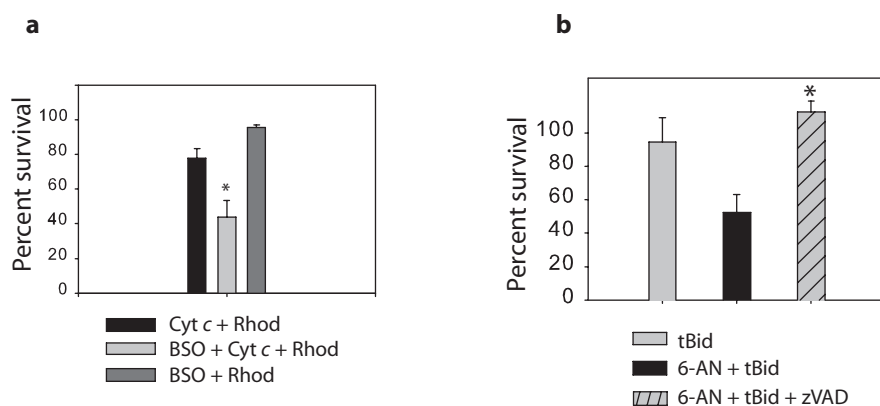


Figure S8 Inhibition of GSH synthesis or the pentose phosphate pathway sensitizes neurons to apoptosis by cytochrome *c*. **a)** XIAP^{-/-} sympathetic neurons were treated with 200 μ M BSO for 16 hrs, followed by injection with cytochrome *c*. Survival was assessed at 3 hrs post-injection. * statistical significance of $n=3$ with $p=.014$ compared to BSO + Rhod. **b)** XIAP^{-/-}

neurons were treated with 6-AN for 24 hrs, followed by injection with tBid along with GFP to mark injected cells. zVAD was added at time of injection. Survival was assessed by quantifying the number of GFP positive cells at 9 hrs vs. 3 hrs post-injection. *statistical significance of $n=3$ with a $p=.017$ as compared to 6-AN + tBid.

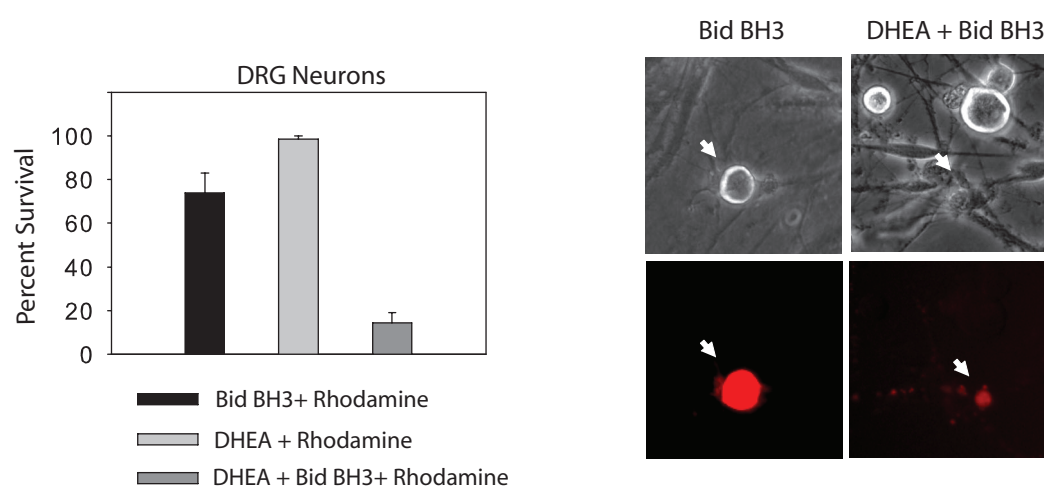


Figure S9 DRG sensory neurons are resistant to cytochrome c release, but become sensitive upon inhibition of the pentose phosphate pathway. Dorsal Root Ganglion neurons were treated with 100 μ M DHEA for 12 hrs, followed

by injection with 2 mM Bid BH3 peptide along with Rhodamine dextran. Cell survival was assessed 6 hrs after injection. Representative photographs are shown. Arrows point to the injected cells. Error bars represent $\pm$ SEM of $n \geq 3$.

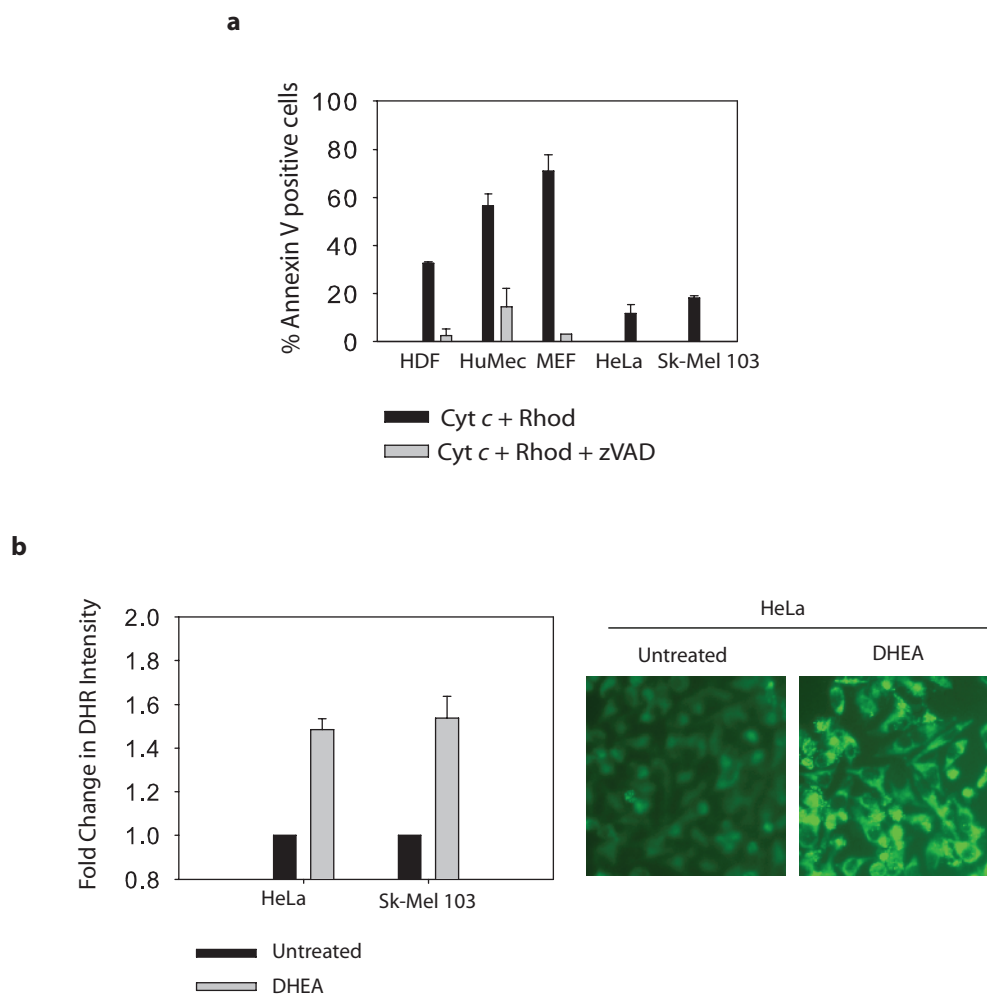


Figure S10 Cytochrome *c* induces apoptosis in normal mitotic cells, and inhibition of the pentose phosphate pathway induces ROS in cancer cells. **a)** Normal mitotic cells were either left untreated, or treated with zVAD, and these cells, in addition to cancer cell lines, were injected with cytochrome *c* (and Smac) along with rhodamine to mark injected cells. One hour following

injections, rhodamine positive cells were assessed for Annexin V-FITC positivity. **b)** HeLa cells and Sk-Mel 103 cells were treated with DHEA for 6 hrs, followed by incubation with 10 μ M Dihydrorhodamine (DHR). Fluorescent intensity of DHR was quantitated on a Fluoroskan plate reader. Representative photographs of HeLa cells are shown. Error bars represent $\pm$ SEM of $n \geq 3$.

Supplementary Methods:

Determination of mitochondrial membrane potential. Prior to fixing to assess cytochrome status, the mitochondrial membrane potential in individual cells was assessed with Mitotracker Orange. Neurons were incubated with 1 μ M Mitotracker (prepared freshly in DMSO) in cell culture medium for 30 min at 37°C in the dark. Neurons were then washed 3 times and individual cells were scored as either highly staining with Mitotracker (retained membrane potential) or weakly staining for Mitotracker (lost membrane potential).

DRG neuronal cultures. Dorsal Root Ganglia were dissected from postnatal day 0, XIAP-deficient C57BL/6 mice and maintained in culture with media containing DMEM, 1% FBS, supplemented with 50 ng/mL NGF. Cells were plated on collagen coated dishes, and grown for 4 days in NGF containing media before treating them with experimental conditions.

Generation and expression of wildtype and N52I Cytochrome *c*. The plasmid pBTR(hCc) (kindly provided by Dr. Gary Pielak, UNC) expresses both the horse cytochrome *c* and yeast heme lyase. A 6 histidine tag was inserted at the N-terminus of pBTR(hCc), and site directed mutagenesis was used to create the N52I mutant. The 6XHis wildtype (WT Cyt *c*) and N52I mutant plasmids were transformed into BL21 (DE3) to express recombinant protein described previously¹. The bacterial pellet was lysed by sonication, and Ni-NTA agarose beads (QIAGEN) were used to purify the His-

tagged protein according to the manufacturer's protocol. The protein was dialyzed overnight, and concentrated with Amicon Ultra 10,000 MWC tubes.

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